Alpha Beta Synthetic Sampling Methods Study

HELIX FORGE LAB / p04-full-005790 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded synthetic sampling evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000015 (models/model-cards/model-reviews/alpha-beta/calibration/tables/alpha-beta-calibration-metrics-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	215	0.892	0.874	0.883
slice-02	177	0.874	0.850	0.862
slice-03	176	0.891	0.863	0.877
slice-04	194	0.873	0.876	0.874
slice-05	174	0.890	0.852	0.871